

PLACEMENTOR

A Minor Project Report

Submitted in partial fulfillment of requirement of the

Degree of

**BACHELOR OF TECHNOLOGY in COMPUTER SCIENCE &
ENGINEERING**

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APRIL 2026

Report Approval

The project work “**Placementor**” is hereby approved as a creditable study of an engineering/computer application subject carried out and presented in a manner satisfactory to warrant its acceptance as prerequisite for the Degree for which it has been submitted.

It is to be understood that by this approval the undersigned do not endorse or approve any statement made, opinion expressed, or conclusion drawn there in; but approve the “Project Report” only for the purpose for which it has been submitted.

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We hereby declare that the project entitled “**Placemtor**” submitted in partial fulfillment for the award of the degree of Bachelor of Technology/Master of Computer Applications in ‘Computer Science Engineering’ completed under the supervision of **Kamini Vishwakarma (Asst. professor, CSE), Priyanka Jain (lecturer, CSE)** Faculty of Engineering, Mediacaps University Indore is an authentic work.

Further, we declare that the content of this Project work, in full or in parts, have neither been taken from any other source nor have been submitted to any other Institute or University for the award of any degree or diploma.

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Certificate

We, **Kamini Vishwakarma, Priyanka Jain** certify that the project entitled **“Placementor”** submitted in partial fulfillment for the award of the degree of Bachelor of Technology by **Ashutosh Mehta, Asmita Gawande, Diksha Garg** is the record carried out by them under our guidance and that the work has not formed the basis of award of any other degree elsewhere.

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Abstract

The rapid evolution of technology and increasing competition in the job market have made placement preparation a challenging task for students. A well-structured resume and awareness of relevant job opportunities play a crucial role in securing employment. However, many students face difficulties in creating effective resumes, understanding Applicant Tracking System (ATS) requirements, and identifying suitable job openings due to lack of proper guidance and integrated tools.

This project, titled “**Placemantor**”, presents an AI-powered placement assistant designed to simplify and enhance the placement preparation process. The system integrates Artificial Intelligence, specifically Large Language Models (LLMs), with web-based technologies to provide an intelligent and automated solution for resume analysis and job discovery. The application enables users to upload resumes in PDF format, which are then processed using a PDF parser to extract textual content. This content is analyzed by an AI agent to evaluate resume quality, generate ATS scores, and provide personalized suggestions for improvement.

A unique feature of the system is the enforcement of GitHub profile inclusion in resumes, encouraging students to showcase their practical skills and projects. Additionally, the system incorporates a job scraping module that collects job and internship opportunities from platforms such as LinkedIn and other job portals. These opportunities are filtered based on user preferences, including location-based filtering through a map interface, ensuring relevant and targeted job recommendations.

The system also includes features such as Google authentication for secure login, multiple resume templates for improved formatting, and a dashboard for tracking user activity and saved jobs. The architecture follows a modular and object-oriented approach, with a Streamlit-based frontend, FastAPI backend, and SQLite database.

Overall, Placemantor provides a comprehensive and user-friendly platform that integrates resume enhancement and job discovery into a single system, thereby improving efficiency, reducing manual effort, and increasing the chances of successful placement.

Keywords: Artificial Intelligence, Resume Analysis, ATS Score, Job Scraping, Placement System, LLM, GitHub Validation

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Abbreviations

This section provides a list of commonly used abbreviations in the report along with their standard meanings. These abbreviations are used to simplify technical terms and improve readability throughout the document.

1. AI – Artificial Intelligence
2. ATS – Applicant Tracking System
3. LLM – Large Language Model
4. API – Application Programming Interface
5. UI – User Interface
6. PDF – Portable Document Format
7. DB – Database
8. SQL – Structured Query Language
9. IDE – Integrated Development Environment
10. NLP – Natural Language Processing
11. HTTP – Hypertext Transfer Protocol
12. JSON – JavaScript Object Notation

Notations & Symbols

1. $f(x)$ – Function representing system operations
2. xxx – Input data (resume content)
3. yyy – Output (analysis results)
4. $\rightarrow$ *|rightarrow* $\rightarrow$ – Data flow direction
5. Σ *|Sigma* Σ – Summation (used in scoring calculations)
6. % – Percentage (used in ATS score representation)
7. ID – Unique Identifier
8. PK – Primary Key
9. FK – Foreign Key
10. $\in$ – Belongs to (used in database relations)

Chapter 1 INTRODUCTION

Introduction

Placement plays a crucial role in shaping the career of students pursuing higher education. In today's competitive job market, students are required to possess not only academic knowledge but also well-structured resumes and awareness of job opportunities. However, many students face challenges in preparing effective resumes and identifying relevant job openings due to lack of guidance and proper tools.

Traditional placement preparation methods involve manual resume building, searching for job opportunities across multiple platforms, and limited feedback on resume quality. These processes are often time-consuming, inefficient, and lack personalization. Students are required to constantly update their resumes and manually check job portals, which leads to missed opportunities and inconsistent preparation.

With the advancement of technology, Artificial Intelligence (AI) has emerged as a powerful tool for automating and enhancing various processes. In particular, Large Language Models (LLMs) have demonstrated significant capability in understanding and generating human-like text, making them suitable for applications such as resume analysis and content improvement.

The proposed system, "Placemantor", is an AI-powered placement assistant designed to assist students in improving their resumes and finding job opportunities efficiently. The system integrates Agentic AI and LLM-based analysis to evaluate resumes, generate ATS (Applicant Tracking System) scores, and provide actionable suggestions for improvement.

The application allows users to upload their resumes in PDF format, which are then processed using a PDF parser to extract textual information. The extracted content is analyzed by an AI agent that identifies strengths, weaknesses, missing keywords, and areas of improvement. Additionally, the system enforces the inclusion of a GitHub profile in the resume to encourage students to showcase their practical projects and technical skills.

Another key feature of the system is job discovery through web scraping. The application collects job and internship opportunities from platforms such as LinkedIn and other job portals. These job listings are filtered based on user preferences, including location selection through a map-based interface, ensuring more relevant results.

The system also provides modern features such as Google authentication for secure login, multiple resume templates for easy resume creation, and a dashboard for tracking history and saved jobs. By combining AI-driven resume analysis with real-time job discovery, Placemantor aims to simplify and enhance the placement preparation process.

Literature Review

Several systems and platforms have been developed to assist students in job search and resume building. Job portals such as LinkedIn, indeed, and Glassdoor provide access to a wide range of job opportunities. These platforms allow users to create profiles, upload resumes, and apply for jobs. However, they primarily focus on job listings and do not provide detailed feedback on resume quality or improvement.

Resume screening systems and Applicant Tracking Systems (ATS) are widely used by companies to filter candidates based on keywords and formatting. While these systems help recruiters manage large volumes of applications, they often reject resumes that are not optimized according to ATS standards. Students are generally unaware of these requirements, which reduces their chances of selection.

AI-based tools for resume analysis have also gained popularity in recent years. These tools use machine learning algorithms to evaluate resumes and suggest improvements. However, many of these systems are limited in functionality and do not provide a comprehensive solution that integrates resume analysis with job discovery.

Research in the field of Agentic AI and Large Language Models has shown that these technologies can be effectively used for intelligent decision-making and text analysis. LLMs are capable of understanding context, identifying patterns, and generating meaningful suggestions, making them suitable for applications in resume evaluation.

Despite the availability of these systems, there exists a gap in providing an integrated platform that combines resume analysis, job discovery, and personalized recommendations. Most existing solutions focus on a single aspect, such as job search or resume building, without offering a unified approach.

The proposed system addresses these limitations by integrating AI-based resume analysis with job scraping and personalized recommendations. It also introduces additional features such as GitHub validation, map-based job filtering, and template-based resume generation, making it a comprehensive solution for placement preparation.

Objectives

- The main objectives of the project are as follows:
- To develop an AI-based system for analyzing resumes.
- To generate ATS scores for evaluating resume quality.
- To provide suggestions for improving resumes.
- To fetch job and internship opportunities using web scraping.
- To provide location-based job filtering using a map interface.
- To implement secure user authentication using Google login.
- To enforce inclusion of GitHub profiles in resumes.
- To provide multiple resume templates for better formatting.
- To assist students in efficient placement preparation.

Significance of the Study

The proposed system is significant as it addresses the common challenges faced by students during placement preparation. It provides an automated and intelligent solution for resume analysis and job discovery, reducing the need for manual effort.

By integrating AI into the system, students receive personalized feedback on their resumes, helping them improve their chances of selection. The inclusion of ATS scoring ensures that resumes are optimized according to industry standards.

The job scraping feature enables students to access opportunities from multiple platforms in a single place, saving time and effort. The map-based filtering enhances usability by allowing users to select preferred job locations easily.

Additionally, features such as GitHub validation encourage students to showcase their practical work, which is an important factor in technical recruitment. Overall, the system contributes to improving placement outcomes and enhancing student preparedness.

Research Design

The project is designed as a software-based system using an object-oriented approach. The development process follows an Agile methodology, allowing iterative improvements and flexibility in implementation.

The system architecture is based on a client-server model. The frontend is developed using Streamlit, which provides an interactive user interface. The backend is implemented using FastAPI, which handles API requests and system logic.

The AI component is built using LangChain integrated with a Large Language Model API, enabling advanced resume analysis and suggestion generation. A PDF parser is used to extract text from uploaded resumes.

The job scraping module is implemented using web scraping techniques to collect job data from platforms such as LinkedIn. The system uses SQLite as a database for storing user data, resumes, and job information.

Source of Data

- The system uses multiple sources of data for its operation:
- User-provided resumes in PDF format.
- Job and internship data collected from LinkedIn and other job portals.
- AI-generated analysis and suggestions based on resume content.
- User input data such as login credentials and location preferences.

Chapter Scheme

The project report is organized into the following chapters:

- **Chapter 1: Introduction** – Provides an overview of the project, literature review, objectives, and significance.
- **Chapter 2: Requirements Specification** – Describes functional and non-functional requirements of the system.
- **Chapter 3: Design** – Presents system design, architecture, and diagrams.
- **Chapter 4: Implementation and Testing** – Explains technologies used and testing methods.
- **Chapter 5: Results and Discussion** – Shows system outputs and performance.
- **Chapter 6: Conclusion** – Summarizes the project and findings.
- **Chapter 7: Future Scope** – Discusses possible improvements and extensions.

Chapter 2 REQUIREMENTS SPECIFICATION

2.1 Introduction

This chapter describes the requirements of the proposed system, Placementor, which is an AI-based placement assistant. It includes functional and non-functional requirements, user characteristics, system constraints, and dependencies.

The purpose of this chapter is to provide a clear understanding of what the system is expected to do and how it should perform under different conditions. These requirements serve as the foundation for system design and implementation.

2.2 Overall Description

Placementor is designed to assist students in improving their resumes and finding relevant job opportunities. The system integrates Artificial Intelligence, web scraping, and database management to provide a comprehensive placement solution.

The application allows users to upload resumes, analyze them using AI, and receive suggestions. It also provides job listings collected from platforms such as LinkedIn and other job portals. Additional features such as Google authentication, GitHub validation, resume templates, and location-based filtering enhance the system functionality.

2.3 User Characteristics

The primary users of the system are students seeking job or internship opportunities.

2.3.1 User Types:

- Students with basic computer knowledge
- Users familiar with uploading documents
- Users with internet access

2.3.2 User Skills:

- Basic understanding of web applications
- Ability to navigate UI
- No advanced technical knowledge required

2.4 Functional Requirements

Functional requirements describe the core features of the system.

2.4.1 User Authentication

- The system shall allow users to register and log in.
- The system shall support Google authentication.
- The system shall validate user credentials.
- The system shall maintain user sessions.

2.4.2 Resume Upload and Processing

- The system shall allow users to upload resumes in PDF format.
- The system shall extract text from resumes using a PDF parser.
- The system shall store resume data in the database.

2.4.3 Resume Validation (GitHub Requirement)

- The system shall verify the presence of a GitHub link in the resume.
- The system shall reject resumes without a GitHub ID.
- The system shall notify users if validation fails.

2.4.4 AI-Based Resume Analysis

- The system shall analyze resume content using AI.
- The system shall generate ATS score.
- The system shall identify strengths and weaknesses.
- The system shall provide improvement suggestions.

2.4.5 Resume Templates

- The system shall provide multiple resume templates.
- The system shall allow users to generate resumes using templates.

2.4.6 Job Scraper System

- The system shall fetch job data from LinkedIn and other platforms.
- The system shall extract job details such as title, company, and location.
- The system shall display job listings to users.

2.4.7 Job Location Map Preference

- The system shall allow users to select job locations using a map.
- The system shall filter job results based on selected location.

2.4.8 Dashboard and History

- The system shall display a user dashboard.
- The system shall maintain user activity history.

2.4.9 Saved Jobs Feature

- The system shall allow users to save job listings.
- The system shall allow users to view saved jobs.

2.5 External Interface Requirements

2.5.1 User Interface

- The system shall provide a simple and user-friendly interface.
- The system shall allow easy navigation between features.
- The interface shall be developed using Streamlit.

2.5.2 Software Interface

- The system shall interact with FastAPI backend.
- The system shall use LLM API for AI processing.
- The system shall integrate Google authentication API.
- The system shall interact with job platforms for scraping.

2.5.3 Hardware Interface

- The system shall run on standard computers or laptops.
- Internet connectivity is required.

2.6 Non-Functional Requirements

2.6.1 Performance Requirements

- The system shall provide fast response time.
- Resume analysis shall complete within a few seconds.
- Job scraping shall be efficient.

2.6.2 Security Requirements

- The system shall ensure secure authentication.
- User data shall be protected.
- API keys shall be securely stored.

2.6.3 Usability Requirements

- The system shall be easy to use.
- Minimal training shall be required.
- Clear feedback messages shall be provided.

2.6.4 Reliability Requirements

- The system shall operate consistently.
- Errors shall be handled properly.
- Resume validation shall be reliable.

2.6.5 Scalability Requirements

- The system shall support multiple users.
- New features can be added easily.

2.6.6 Availability Requirements

- The system shall be available whenever required.
- External services such as Google login shall be accessible.

2.6.7 Maintainability Requirements

- The system shall follow modular design.
- Code shall be easy to update and maintain.

2.7 Constraints and Assumptions

2.7.1 Constraints

- Internet connection is required.
- System depends on external APIs.
- Web scraping depends on website structure.

2.7.2 Assumptions

- Users have basic knowledge of system usage.
- Resume is uploaded in correct format.
- External APIs function properly.

2.8 Dependencies

- Dependency on LLM API for analysis
- Dependency on LinkedIn and job portals for data
- Dependency on Google authentication service

Chapter 3: SYSTEM DESIGN

3.1 Introduction

System design is a crucial phase in software development that defines the architecture, components, and interactions within the system. It provides a blueprint for implementing the system and ensures that all requirements are met efficiently.

The Placemtor system is designed using a modular and object-oriented approach. It integrates various components such as frontend, backend, AI modules, and database to provide a seamless user experience. The system is structured to handle resume analysis, job scraping, and user interaction effectively.

3.2 System Design Overview

The Placemtor system follows a client-server architecture. The frontend is responsible for user interaction, while the backend processes requests and manages system logic. The system also includes AI-based modules and external integrations.

The main components of the system are:

- Frontend (Streamlit)
- Backend (FastAPI)
- AI Agent (LangChain + LLM API)
- Resume Processing Module
- Job Scraper Module
- Authentication Module (Google + Manual Login)
- Database (SQLite)

These components interact with each other to provide functionalities such as resume analysis, job recommendations, and user management.

3.3 Data Flow Diagrams

3.3.1 Level 0 DFD

The Data Flow Diagram (Level 0) provides a high-level overview of the entire system. It shows the Placemotor system as a single process interacting with external entities such as the user and job providers. It represents the flow of data such as login information, resume files, job requests, and results between the user and the system.

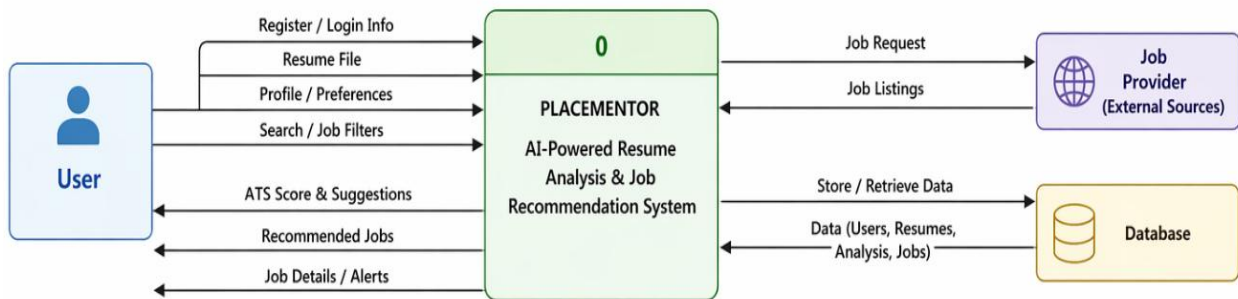


fig3.1: dfd level 0

3.3.2 Level 1 DFD

The DFD Level 1 breaks down the system into detailed processes such as user authentication, resume processing, AI analysis, and job fetching. It shows how data flows between these processes and the database. This diagram provides a deeper understanding of how the system internally handles operations step-by-step.

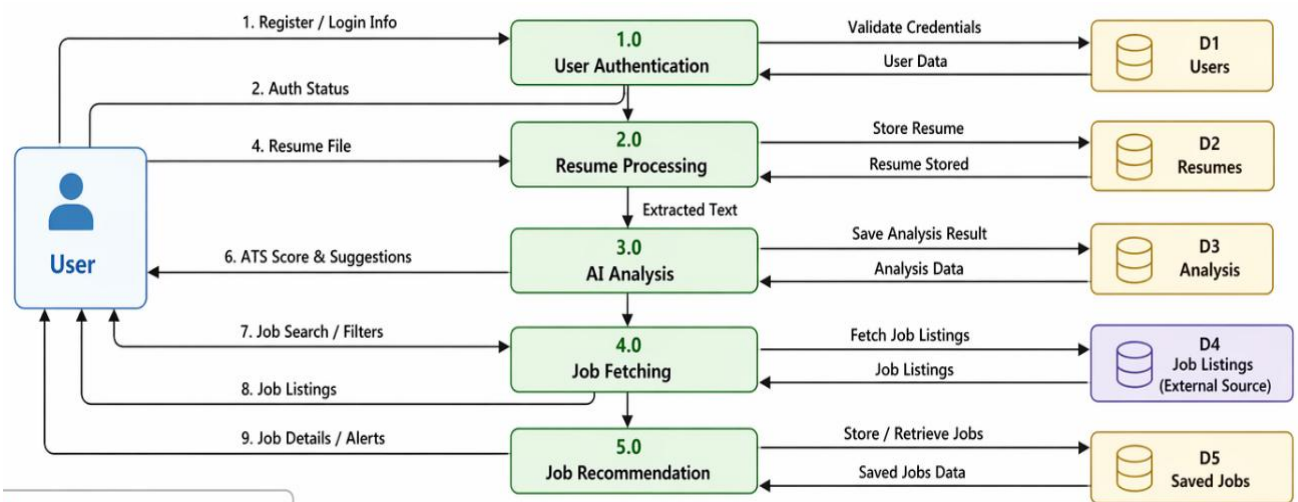


fig3.2: dfd level 1

3.4 Activity Diagram

The Activity Diagram represents the workflow of the system from start to end. It shows the sequence of actions such as user login, resume upload, text extraction, AI analysis, job search, and result display. It also includes decision points such as authentication validation. This diagram helps in understanding the overall process flow of the system.

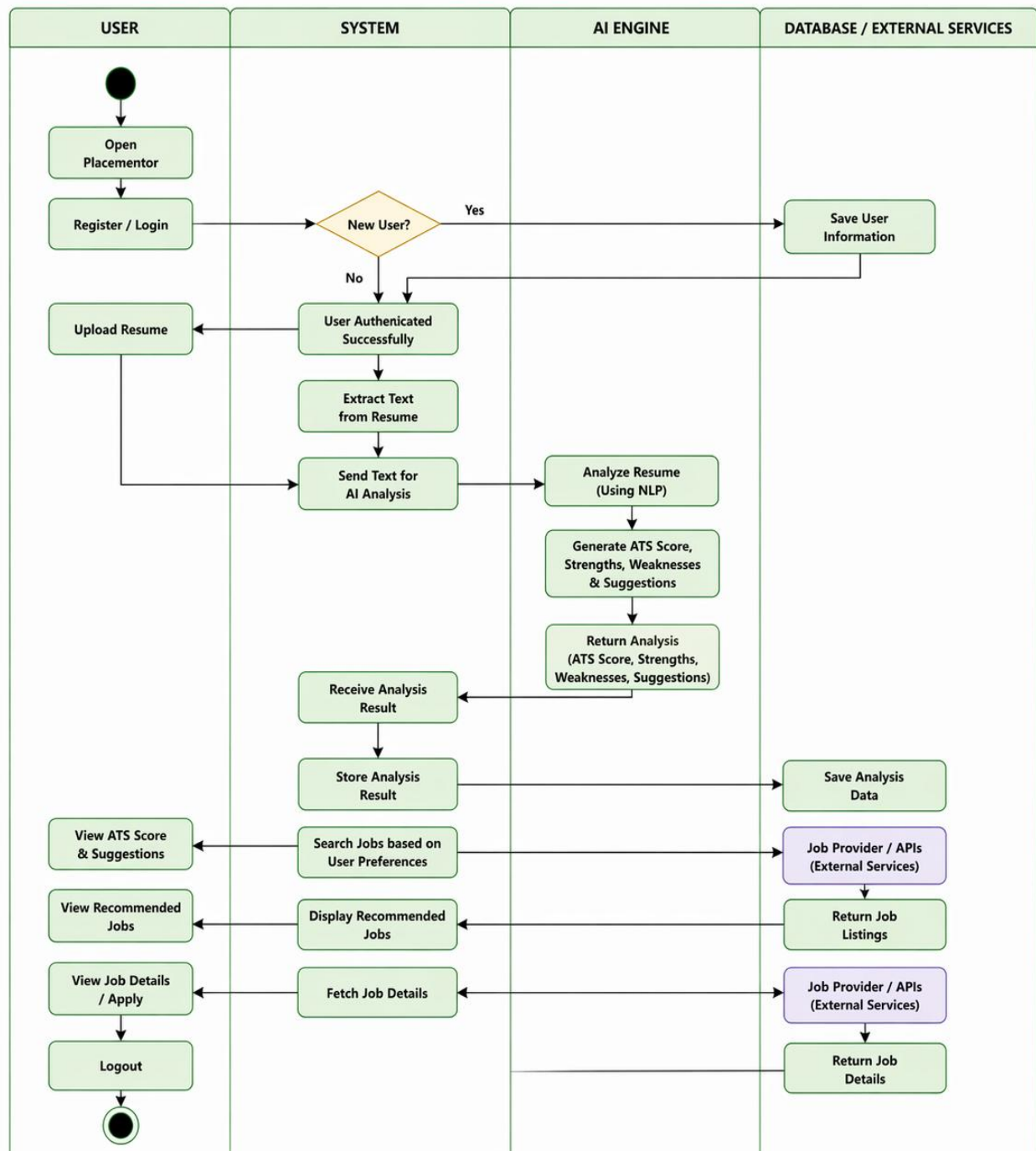


fig3.3: activity diagram

3.5 Flowchart

The Flowchart provides a step-by-step representation of the system process. It shows how the system starts from user login, processes the resume, performs AI analysis, fetches job data, and displays results. It includes decision-making steps such as authentication success or failure. This diagram helps in understanding the logical flow of the system.

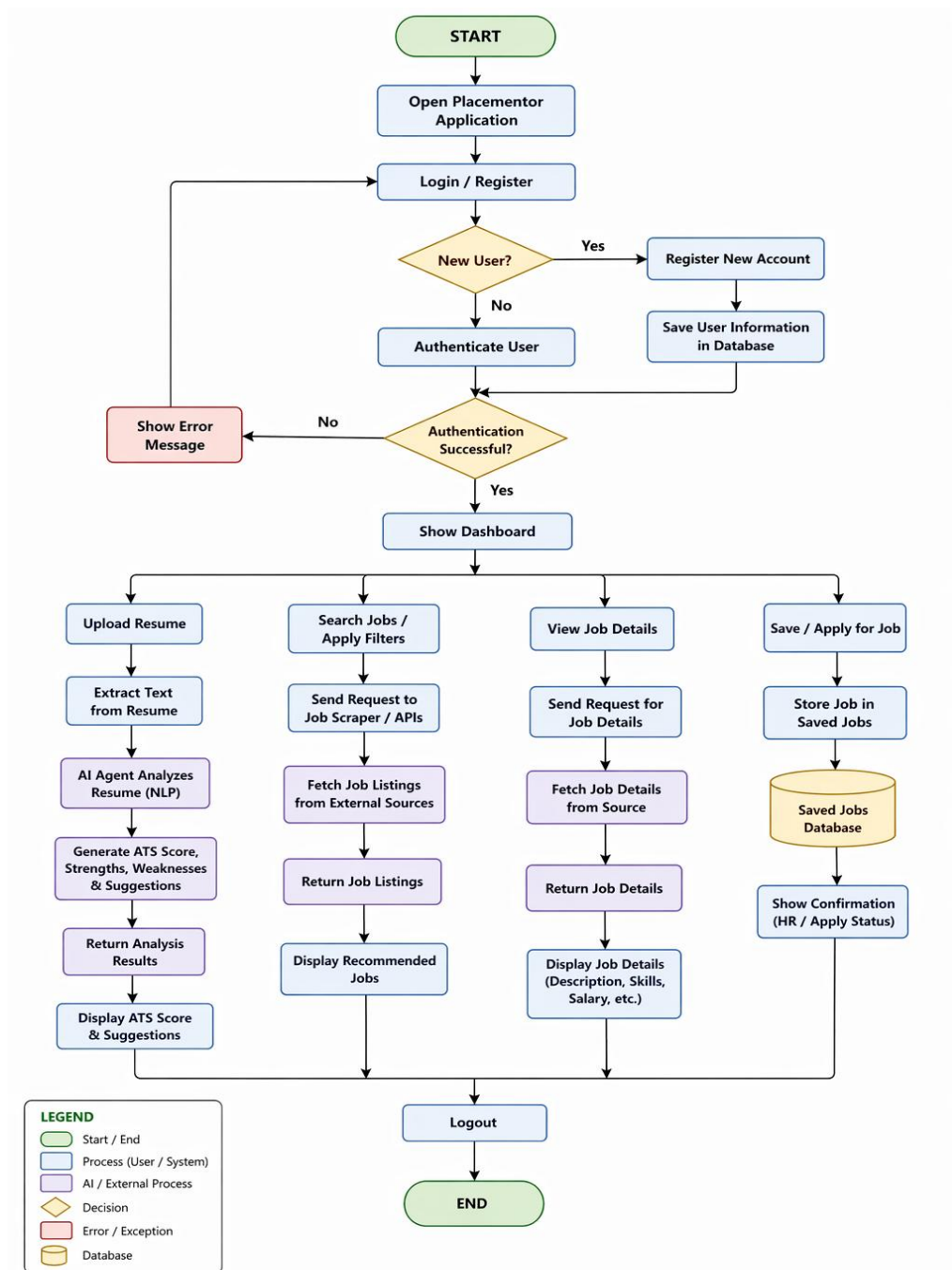


fig3.4: flowchart

3.6 Class Diagram

The Class Diagram illustrates the structural design of the system by showing different classes such as User, Resume, Analysis, Job, AI Agent, Job Scraper, and Database. It defines the attributes and functions of each class and shows the relationships between them. This diagram helps in understanding how data is organized and how different components interact within the system.

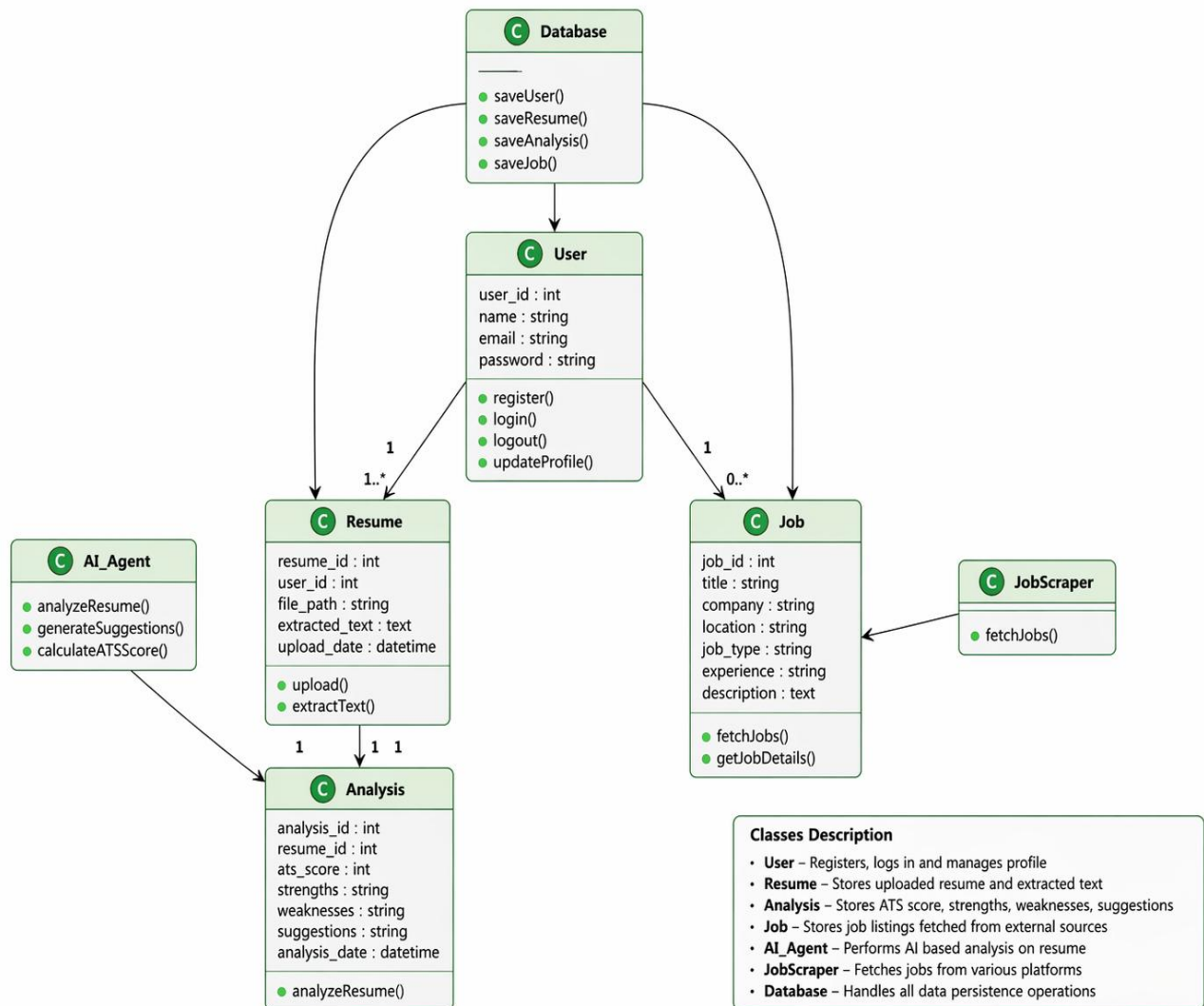


fig3.5: class diagram

3.7 ER Diagram

The Entity Relationship (ER) Diagram represents the database structure of the Placementor system. It shows entities such as User, Resume, Analysis, Job, and Saved Jobs along with their attributes. The relationships between these entities, such as a user uploading resumes and generating analysis results, are clearly defined. This diagram helps in understanding how data is stored and managed in the database.

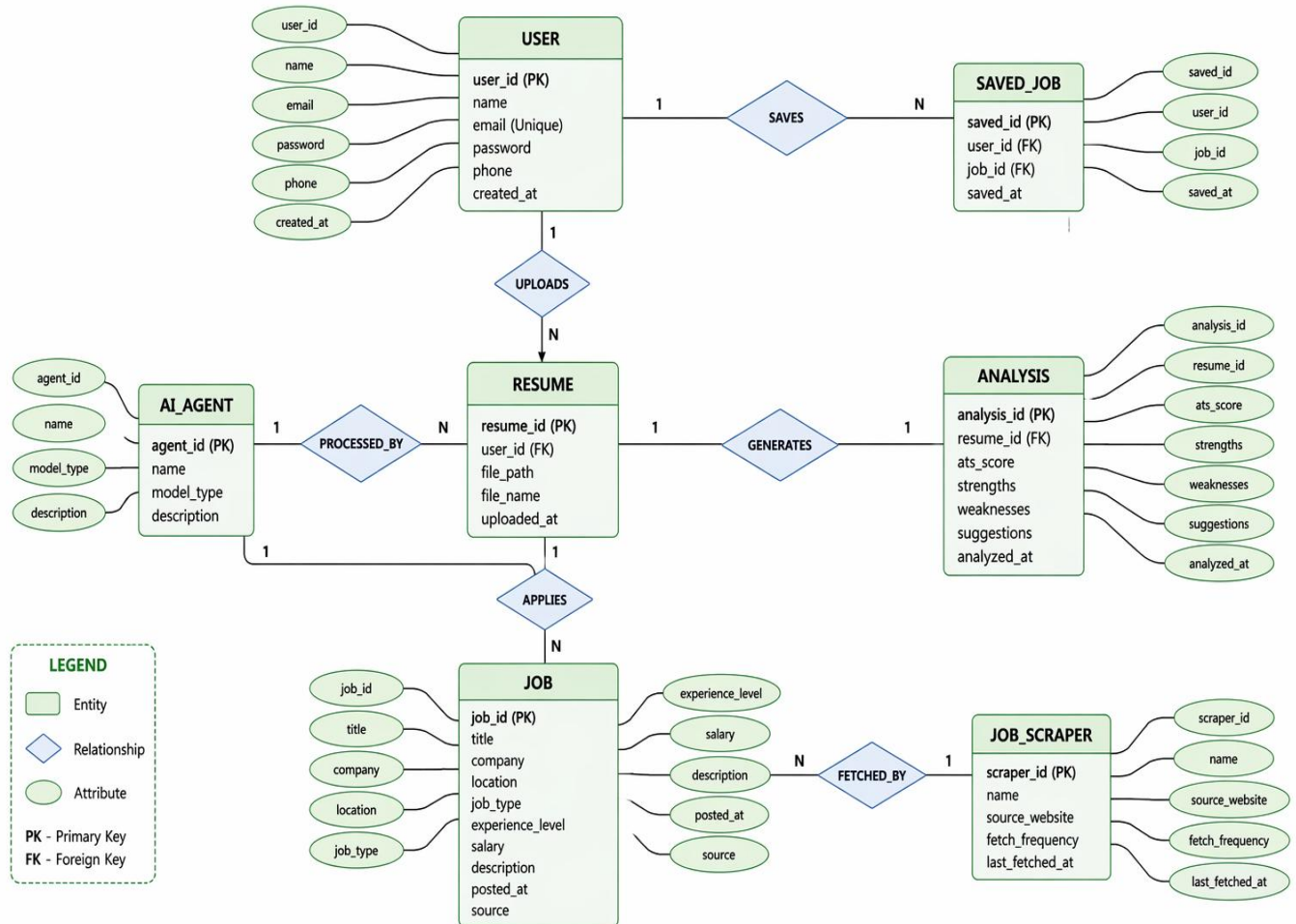


fig3.6: ER diagram

3.8 Sequence Diagram

The Sequence Diagram illustrates the interaction between different components of the system over time. It shows how the user, frontend, backend, AI agent, database, and job scraper communicate with each other. The diagram represents the sequence of messages exchanged during operations such as login, resume analysis, and job fetching.

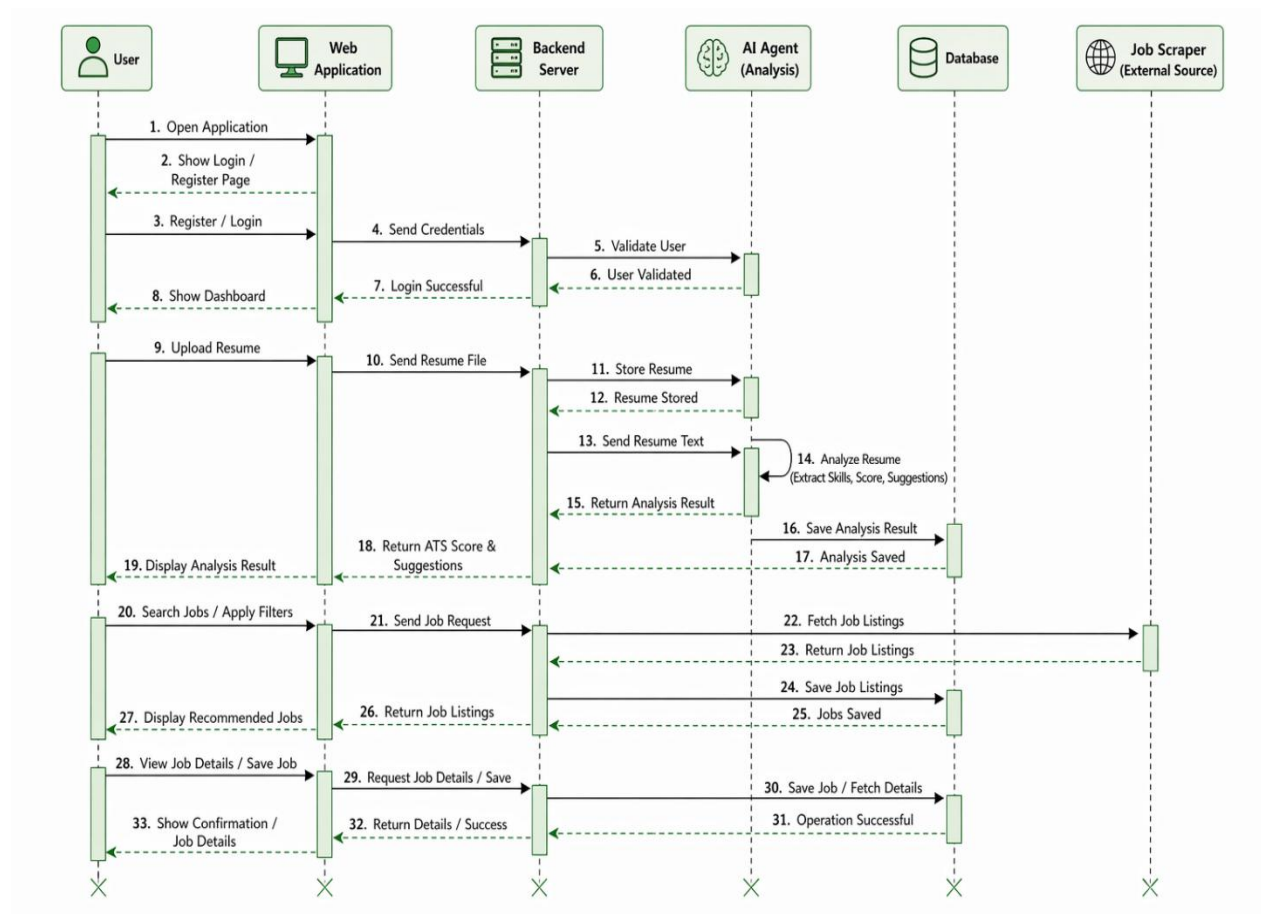


fig3.7: sequence diagram

3.9 Use Case Diagram

The Use Case Diagram represents the interaction between the user and the Placemotor system. It shows the different functionalities available to the user such as registration, login, resume upload, resume analysis, job search, and viewing job details. The diagram also highlights how the system internally processes these actions using AI analysis and job scraping services. It provides a high-level understanding of system functionalities from the user's perspective.

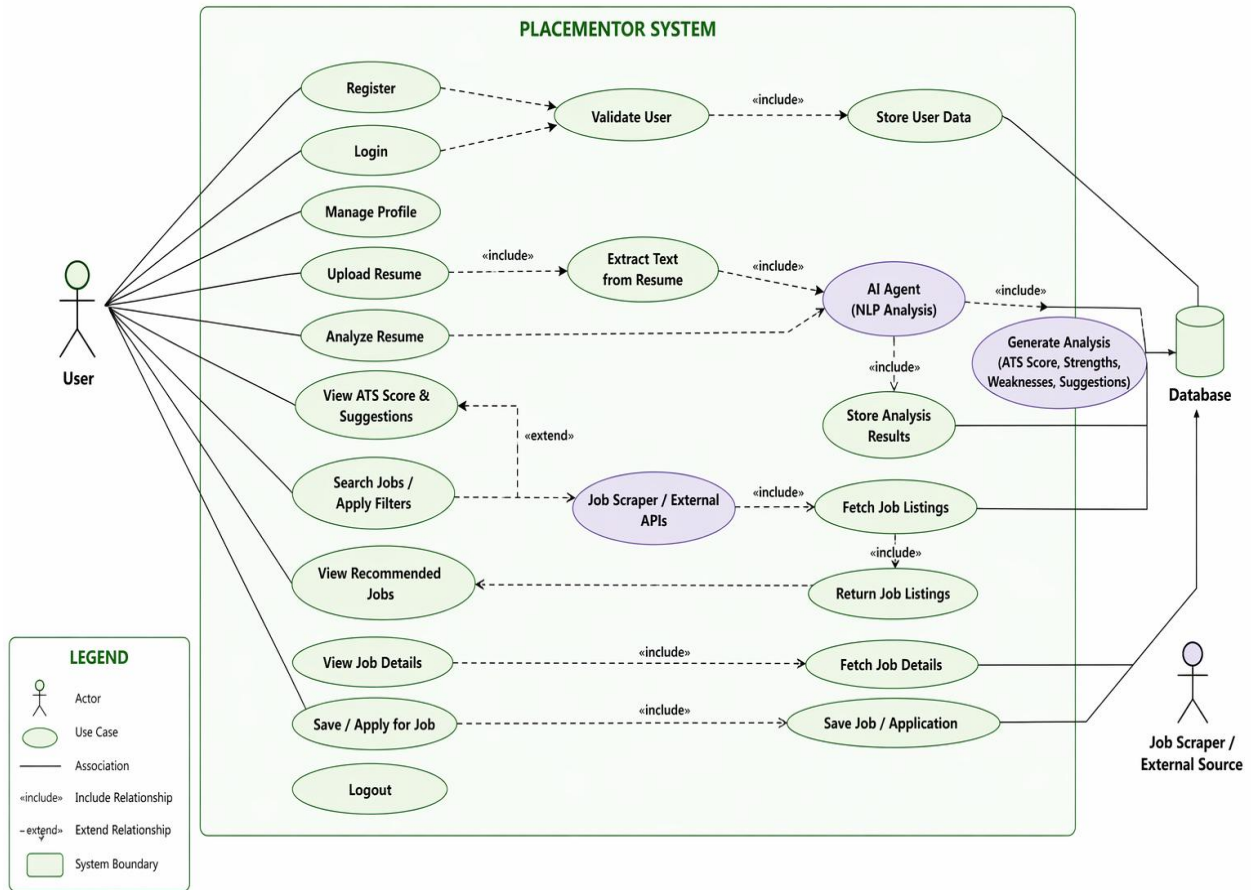


fig3.8: use case diagram

3.10 Database Design

3.10.1 Logical Database Design

The logical database design defines the structure of the database in terms of tables and relationships.

Main tables include:

- User Table (user_id, email, password)
- Resume Table (resume_id, user_id, file, extracted_text)
- Analysis Table (analysis_id, resume_id, ATS score, suggestions)
- Job Table (job_id, title, company, location)
- Saved Jobs Table (user_id, job_id)

3.10.2 Physical Database Design

The physical database is implemented using SQLite. The database stores user data, resumes, analysis results, and job information efficiently.

Indexes and keys are used to improve performance and ensure data integrity.

3.11 System Architecture

The system architecture follows a modular design approach:

- The frontend provides user interface for interaction.
- The backend handles API requests and business logic.
- The AI agent processes resume data and generates insights.
- The job scraper collects job data from external sources such as LinkedIn.
- Google authentication is used for secure login.
- The database stores all system data.

This architecture ensures scalability, maintainability, and efficient processing.

3.12 Module Design

The system is divided into the following modules:

3.12.1 Authentication Module

- Handles user login and registration
- Supports Google authentication
- Manages user sessions

3.12.2 Resume Processing Module

- Handles resume upload
- Extracts text using PDF parser
- Validates GitHub link

3.12.3 AI Analysis Module

- Uses LLM to analyze resumes
- Generates ATS score
- Provides suggestions

3.12.4 Job Scraper Module

- Fetches job data from LinkedIn and other platforms
- Filters jobs based on location

3.12.5 Dashboard Module

- Displays analysis results
- Shows job listings
- Maintains history and saved jobs

3.13 Summary

This chapter presented the overall design of the Placementor system, including system architecture, diagrams, and database structure. The modular design ensures efficient implementation and easy scalability.

CHAPTER 4: IMPLEMENTATION, TESTING AND MAINTENANCE

4.1 Introduction

This chapter describes the implementation details of the Placemotor system, including the technologies used, development process, testing methodologies, and maintenance strategies. The implementation phase is one of the most critical stages of the software development lifecycle, where the system design is translated into a functional application.

The Placemotor system is developed using a modular and scalable architecture. Each component of the system is implemented independently to ensure flexibility, maintainability, and ease of future enhancements. The system integrates Artificial Intelligence, web scraping, and database management to provide an intelligent placement assistance platform.

The implementation focuses on achieving the objectives defined in earlier chapters, including resume analysis using AI, job discovery through scraping, and user-friendly interaction through a web interface.

4.2 Technologies, Tools and Platforms Used

The system is built using modern technologies that ensure efficiency, scalability, and performance.

4.2.1 Frontend Technology (Streamlit)

Streamlit is used to develop the frontend of the application. It provides a simple yet powerful interface for building interactive web applications using Python.

- Enables rapid development of user interfaces
- Supports widgets such as buttons, forms, and file uploads
- Provides real-time updates and visualization
- Used to create dashboard, resume upload page, and job listing interface

4.2.3 Database (SQLite)

SQLite is used as a lightweight database for storing system data.

- Stores user information, resumes, and job data
- Provides efficient data retrieval and storage
- Suitable for small to medium-scale applications
- Ensures data consistency and integrity

4.2.4 Artificial Intelligence Technologies

The system integrates AI using:

- **LangChain:** Used to build intelligent agents
- **LLM API:** Used for natural language processing

The AI module performs:

- Resume analysis
- Keyword detection
- Suggestion generation
- ATS scoring

4.2.5 Additional Tools and Technologies

- **PDF Parser:** Extracts text from uploaded resumes
- **Web Scraping Tools (BeautifulSoup):** Collect job data
- **Google Authentication API:** Enables secure login
- **Map Interface:** Used for location-based job filtering

4.3 System Implementation

The system is implemented in a modular structure where each module performs a specific function.

4.3.1 Authentication Module

The authentication module is responsible for verifying user identity and providing secure access to the system.

- Supports manual login using email and password
- Supports Google authentication for faster login
- Ensures secure session management
- Prevents unauthorized access

4.3.2 Resume Processing Module

This module handles resume upload and preprocessing.

- Accepts resumes in PDF format
- Extracts text using PDF parser
- Validates the presence of GitHub link
- Stores resume data in database

This validation ensures that users include technical work in their resumes.

4.3.3 AI Analysis Module

The AI module is the core component of the system.

- Processes extracted resume text
- Generates ATS score
- Identifies strengths and weaknesses
- Suggests improvements

The module uses LLM capabilities to understand context and provide meaningful insights.

4.3.4 Job Scraper Module

This module collects job data from external platforms.

- Scrapes job listings from LinkedIn and other sites
- Extracts job details such as title, company, and location
- Filters jobs based on user preferences

4.3.5 Location Filtering Module

- Allows users to select job locations using a map
- Filters job results based on selected location
- Improves relevance of job recommendations

4.3.6 Resume Template Module

- Provides multiple resume templates
- Allows users to generate structured resumes
- Helps improve formatting and presentation

4.3.7 Dashboard Module

- Displays analysis results and job listings
- Maintains user history
- Allows saving of jobs

4.4 Testing Techniques and Test Plan

Testing is performed to ensure system reliability and correctness.

4.4.1 Testing Objectives

- Verify system functionality
- Detect errors and bugs
- Ensure smooth user experience

4.4.2 Types of Testing

Unit Testing

- Individual modules tested separately
- Example: Resume upload, AI analysis

Integration Testing

- Interaction between modules tested
- Example: Frontend + backend + database

System Testing

- Entire system tested as a whole
- End-to-end functionality verified

Manual Testing

- System tested manually by users
- Real-world scenarios executed

4.4.3 Detailed Test Cases

Test Case	Input	Expected Output
Login	Valid credentials	Login successful
Google Login	Google account	Access granted
Resume Upload	PDF file	Text extracted
GitHub Check	Missing GitHub	Error shown
AI Analysis	Resume text	ATS score
Job Scraping	Request	Jobs displayed
Location Filter	Selected location	Filtered jobs

Table 4.1: test cases table

4.5 Installation Instructions

The Placementor system can be installed and executed on any system that supports Python and required libraries. The following steps describe the setup process.

4.5.1 System Requirements

- Operating System: Windows/Linux/macOS
- Python: Version 3.8 or above
- RAM: Minimum 4 GB
- Internet connection for API and job scraping

4.5.2 Setup Process

1. Download or clone the project files.
2. Open the project folder in a code editor.
3. Install required dependencies using:
`pip install -r requirements.txt`

4.5.3 Configuration

- Create a .env file for API keys.
- Add required keys such as LLM API key and Google authentication credentials.

4.5.5 Accessing the System

- Open browser and go to <http://localhost:8501>
- Login using credentials or Google account
- Upload resume and use system features

4.5.6 Verification

- Check login functionality
- Upload resume and verify analysis
- Ensure job listings are displayed

4.6 End User Instructions

The Placemator system is designed to be simple and user-friendly. The following steps guide the user in operating the application:

- Open the application in a web browser.
- Login using email/password or Google authentication.
- Upload the resume in PDF format.
- Ensure the resume contains a GitHub link for validation.
- View ATS score and suggestions generated by the system.
- Browse job listings fetched from online platforms.
- Use the map to select preferred job locations.
- Save jobs for future reference and view history in the dashboard.

4.7 Maintenance

Maintenance of the system ensures smooth functioning and continuous improvement. The following types of maintenance are considered:

- **Corrective Maintenance:** Fix bugs and errors identified during usage.
- **Adaptive Maintenance:** Update the system to meet new requirements or changes in external platforms.
- **Perfective Maintenance:** Improve system performance and add new features.
- **Preventive Maintenance:** Take measures to avoid future issues and ensure system stability.

Regular updates to AI models, job scraping logic, and system components help maintain accuracy and efficiency.

4.8 Summary

This chapter presented the implementation details of the Placementor system, including the technologies used, system modules, and overall working process. It also described the testing techniques used to verify system functionality and ensure reliability.

The installation steps and user instructions were provided to guide both developers and end users in using the system effectively. Additionally, maintenance strategies were discussed to ensure long-term performance and scalability of the application.

Overall, the implementation and testing confirm that the system functions correctly and meets the intended objectives of assisting students in resume analysis and job discovery.

CHAPTER 5: RESULTS AND DISCUSSION

5.1 Introduction

This chapter presents the results obtained from the implementation of the Placementor system. It includes system outputs, screenshots, and analysis of different modules. The purpose of this chapter is to demonstrate the effectiveness and functionality of the developed system.

The system has been tested under various scenarios to ensure correct performance of features such as resume analysis, job scraping, authentication, and dashboard functionality.

5.2 User Interface Representation

The Placementor system provides a simple and interactive user interface built using Streamlit. The interface allows users to easily interact with the system and access various features.

5.2.1 Login Page

Description:

The login page allows users to access the system using either manual login or Google authentication. It ensures secure access and user validation.

5.2.2 Resume Upload Page

Description:

Users can upload resumes in PDF format. The system checks for the presence of a GitHub link and validates the resume before processing.

5.2.3 GitHub Validation Output

Description:

The system includes a GitHub validation feature that checks whether the uploaded resume contains a valid GitHub profile link. If the link is missing, the system displays an error message prompting the user to include it. This ensures that users showcase their project work and technical skills.

5.2.3 Location-Based Filtering

Description:

The system provides job filtering based on user-selected location. Users can specify their preferred location, and the system displays relevant job opportunities accordingly. This feature helps in improving the accuracy of job recommendations.

5.2.3 Resume Analysis Output

Description:

The system displays ATS score along with strengths, weaknesses, and suggestions for improvement. This helps users enhance their resumes effectively.

5.2.4 Job Listings Page

Description:

This page displays job opportunities fetched from LinkedIn and other platforms. Users can filter jobs based on location using a map-based interface.

5.2.5 Dashboard and Saved Jobs

Description:

The dashboard shows user activity, saved jobs, and previous resume analysis results, providing a centralized view of all data.

Screen Name	Description
Login Page	Allows users to log in using credentials or Google authentication
Resume Upload	Enables users to upload resumes in PDF format
GitHub Validation	Checks presence of GitHub link in resume
Resume Analysis	Displays ATS score and improvement suggestions
Job Listings	Shows jobs fetched from LinkedIn and other platforms
Dashboard	Displays saved jobs and user activity

Table 5.1 : System Screens Description

5.3 Description of System Modules

The system is divided into several modules, each responsible for specific functionality.

5.3.1 Authentication Module

- Successfully supports manual and Google login
- Secure user validation
- Smooth session handling

5.3.2 Resume Processing Module

- Successfully uploads PDF resumes
- Extracts text accurately
- Validates GitHub link correctly

5.3.3 AI Analysis Module

- Generates ATS score
- Provides meaningful suggestions
- Identifies missing keywords

5.3.4 Job Scraper Module

- Fetches job listings from LinkedIn
- Displays relevant opportunities
- Supports location-based filtering

5.3.5 Dashboard Module

- Displays user history
- Allows saving jobs
- Provides easy navigation

Module	Functionality	Result
Authentication	Login & signup	Successful
Resume Processing	Upload & parsing	Accurate
GitHub Validation	Link checking	Works correctly
AI Analysis	ATS score generation	Effective
Job Scraper	Fetch jobs	Relevant results
Dashboard	Display history	Functional

Table 5.2: System Modules Table

5.4 Output of the System

5.4.1: Login Page

The login page allows users to securely access the system using their credentials or Google authentication. It ensures that only authorized users can use the application features.

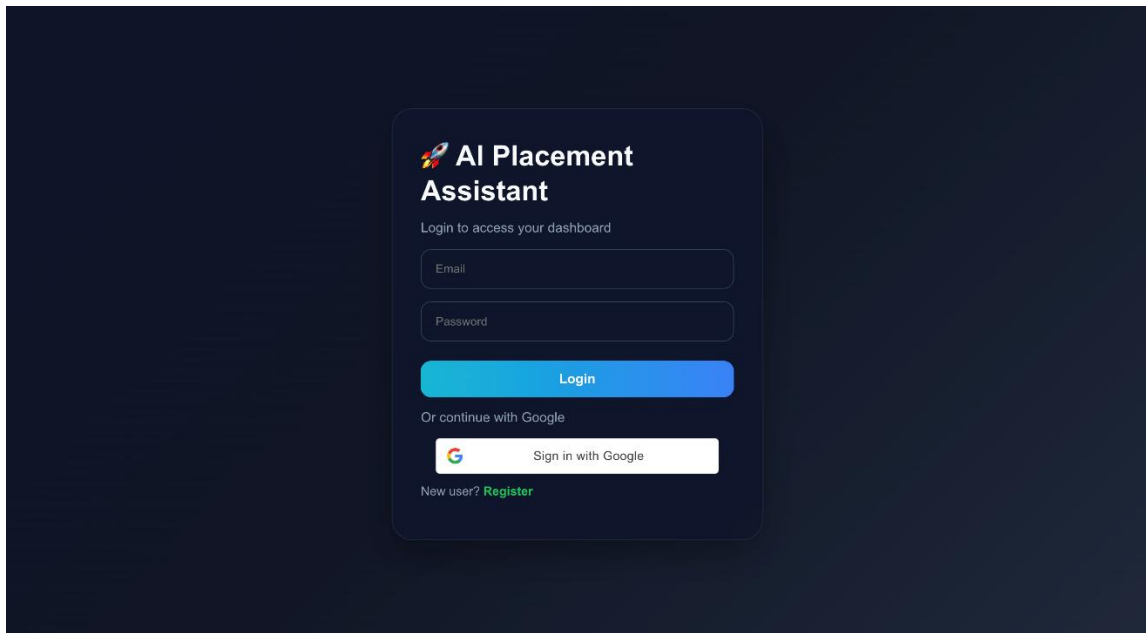


fig5.1: login page

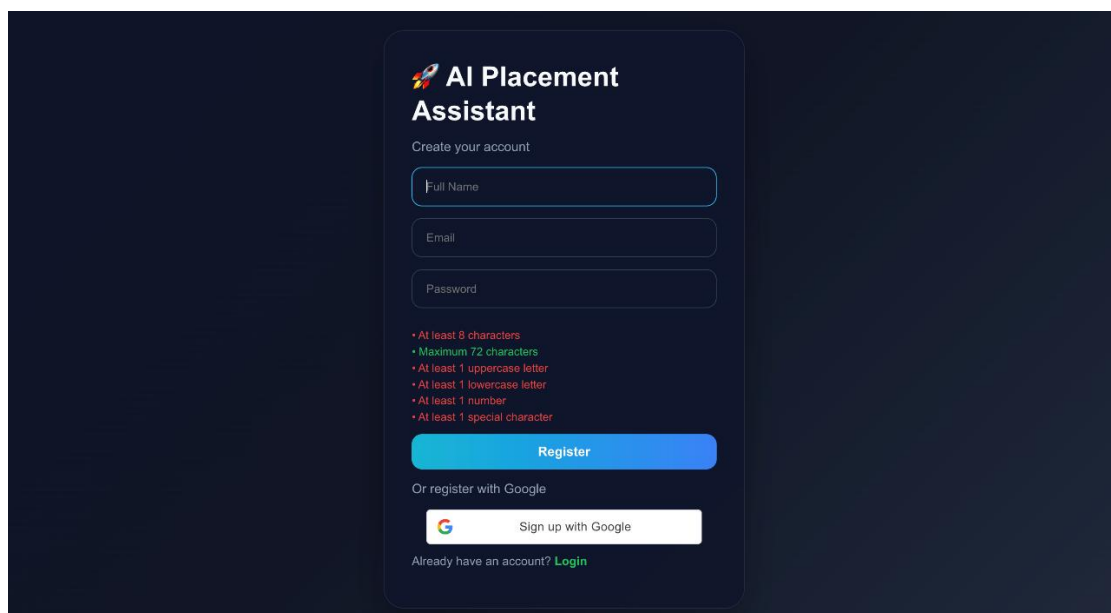


fig5.2: new user registration

5.4.2: Dashboard

The dashboard provides an overview of the system features and user activities. It acts as the central interface from where users can navigate to resume upload, job search, and analysis sections.

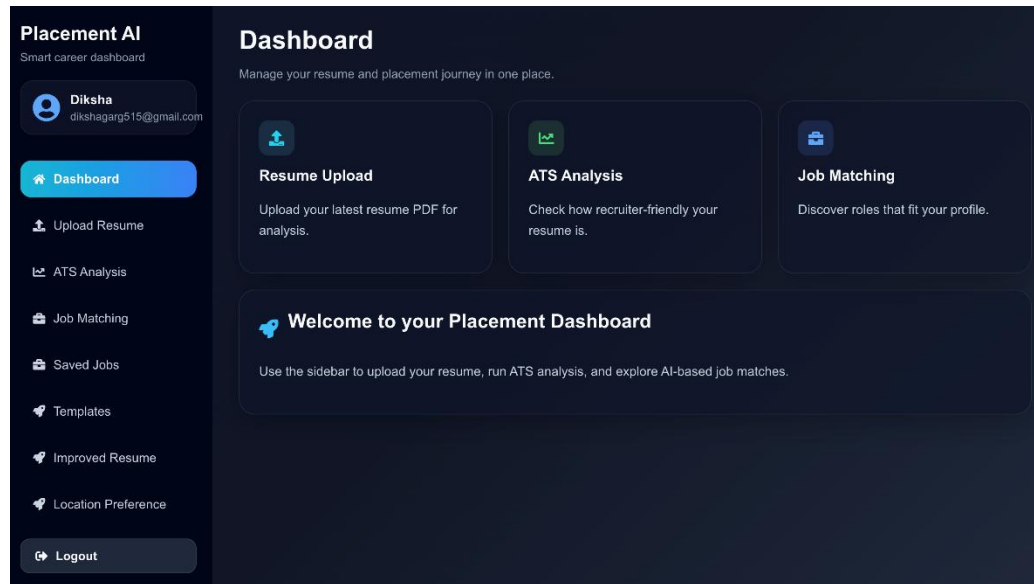


fig5.3: dashboard

5.4.3: Resume Upload Page

This page allows users to upload their resumes in PDF format. The system extracts text from the uploaded resume and prepares it for further analysis.

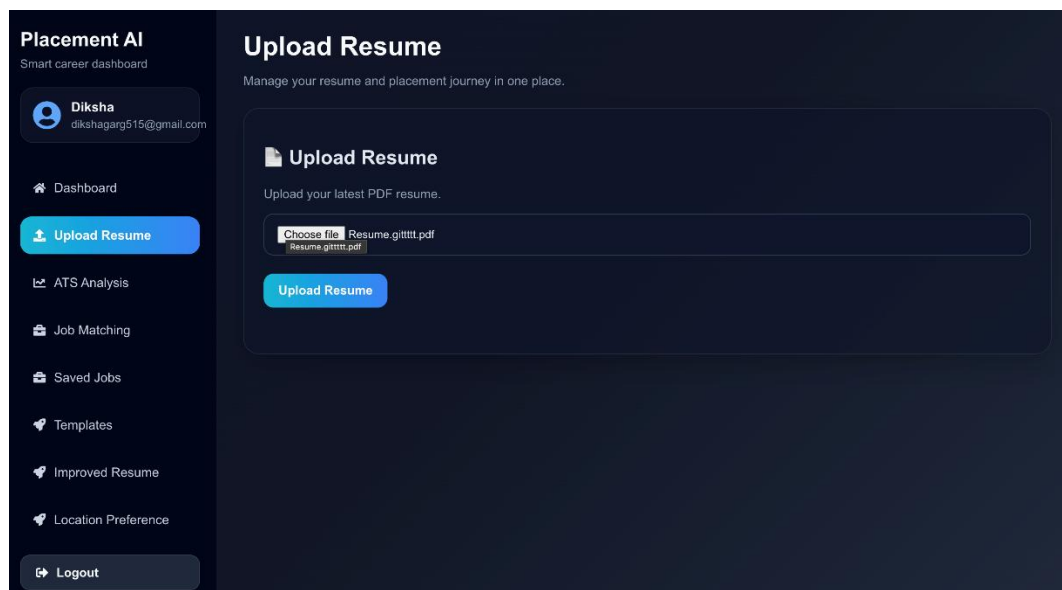


fig5.4: resume upload page

5.4.4: Resume Template Selection

This feature allows users to choose from different resume templates. It helps users create well-structured and professional resumes based on predefined formats.

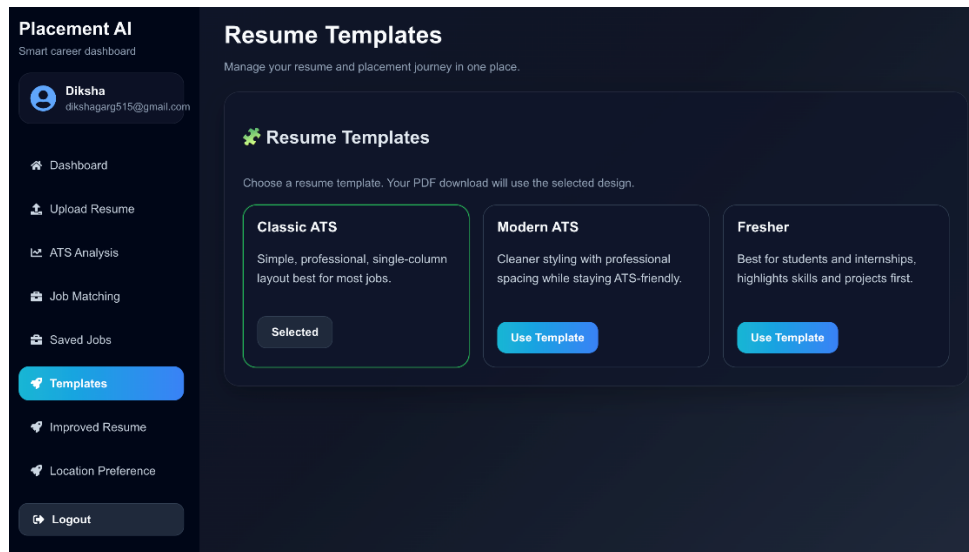


fig5.5: resume templates

5.4.5: Improved Resume Output

The system generates an improved version of the resume based on AI suggestions. It enhances formatting, content quality, and keyword optimization.

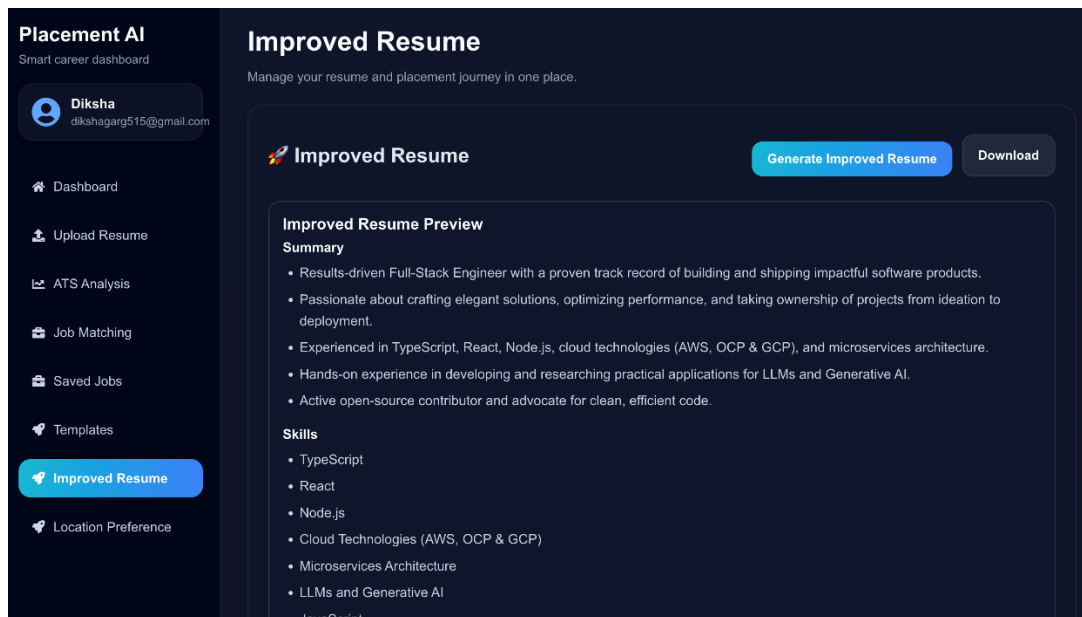


fig5.6: improved resume suggestion

5.4.5: ATS Score and Analysis Output

The system generates an ATS score based on the resume content. It also provides suggestions, strengths, weaknesses, and missing keywords to improve the resume.

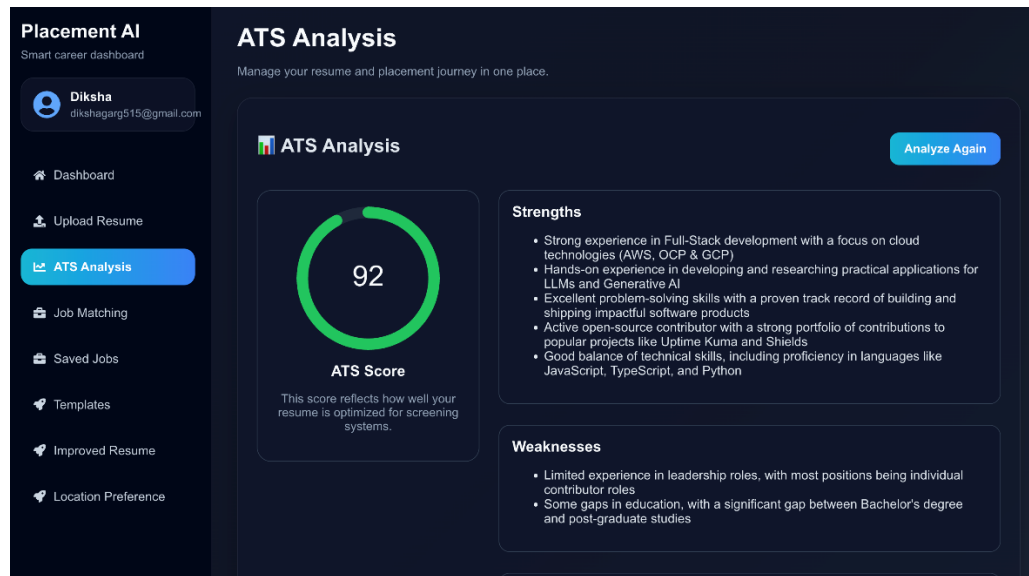


fig5.7: ATS analysis

5.4.8: Location-Based Job Filtering

This feature allows users to filter job results based on preferred locations. It improves job relevance and helps users find opportunities in specific regions.

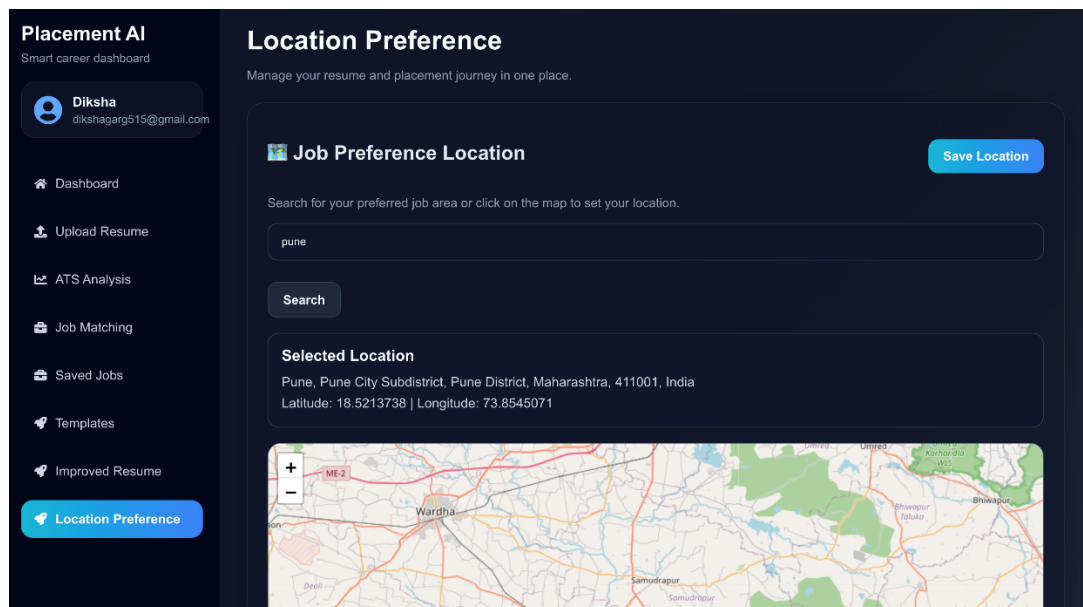


fig5.8: location based filtering

5.4.7: Job Listings Page

This page displays job opportunities fetched from LinkedIn and other platforms. The jobs are relevant to the user's resume and skills.

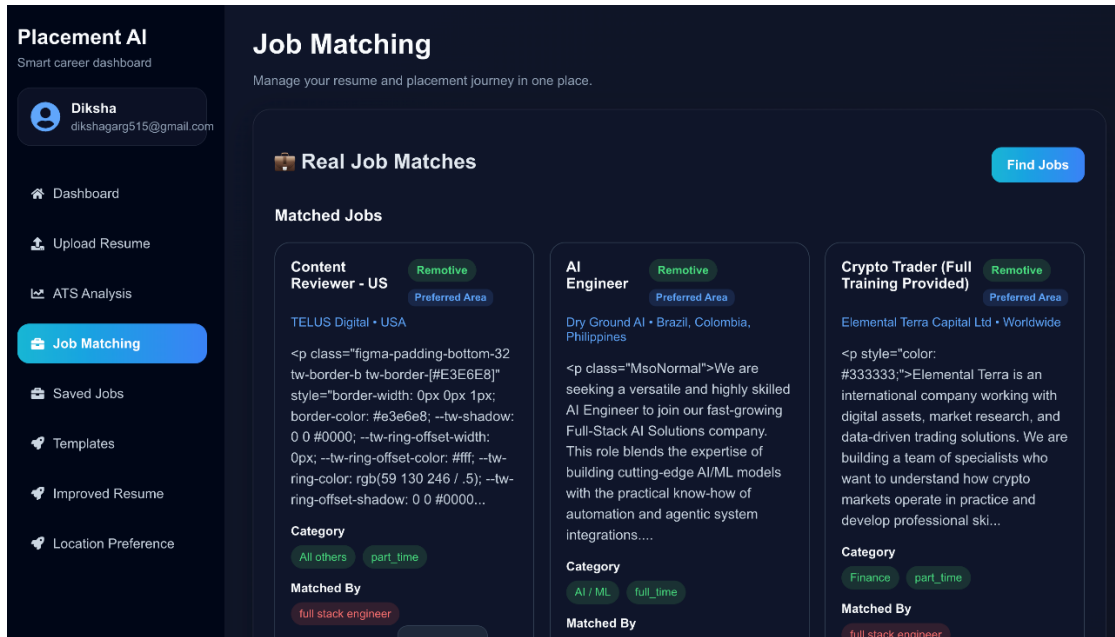


fig5.9: job matching

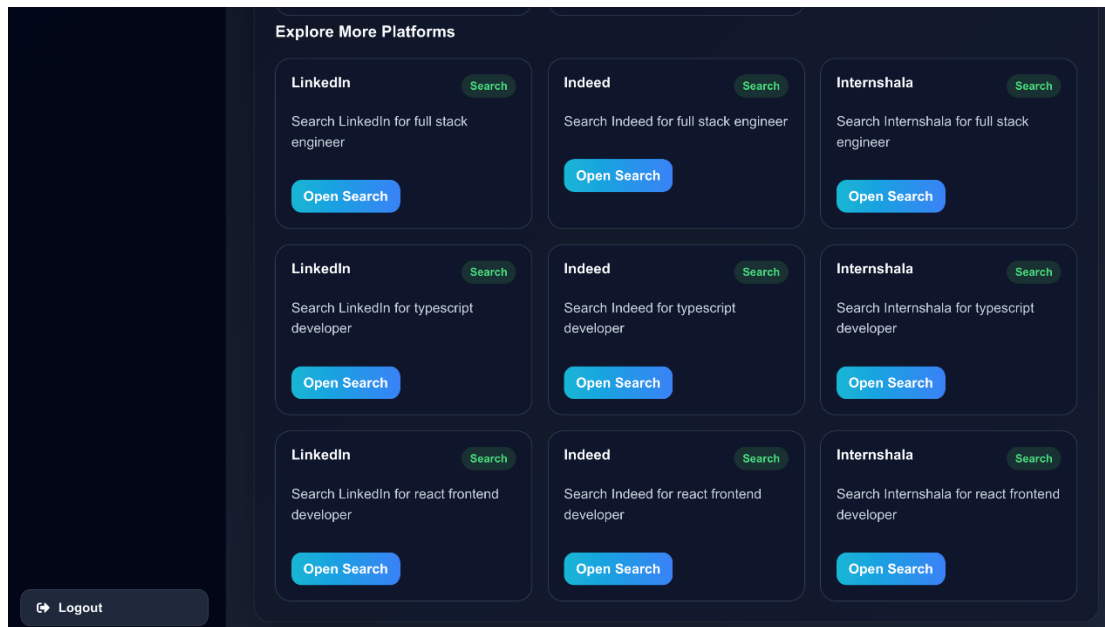


fig5.10: jobs on more platform

Figure 5.4.8: Saved Jobs / History

This section allows users to save job listings and view their previous activities such as analyzed resumes and results.

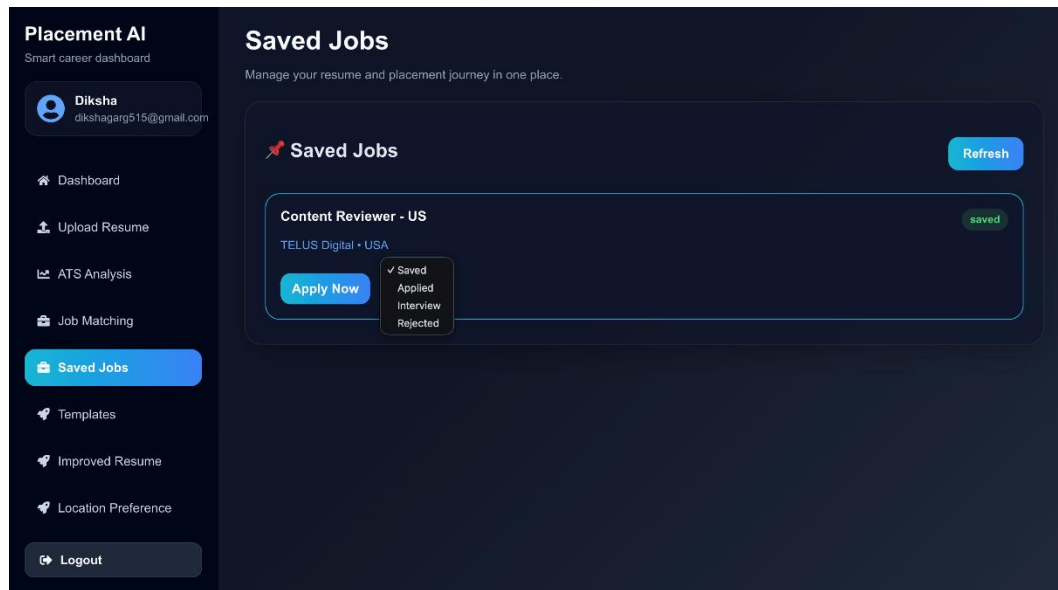


fig5.11: saved jobs

5.5 Performance Analysis

The system performs efficiently under normal conditions. Resume processing and AI analysis are completed within a few seconds. Job scraping takes slightly more time depending on internet connectivity.

The database ensures efficient storage and retrieval of data. The modular architecture allows smooth communication between components.

Parameter	Observation
Resume Processing Time	Few seconds
AI Analysis Speed	Fast response
Job Scraping Speed	Moderate (depends on internet)
System Response	Smooth
Database Performance	Efficient

Table 5.3: Performance table

Test Case	Input	Expected Output	Result
Login	Valid user	Login success	Passed
Resume Upload	PDF	Processed	Passed
AI Analysis	Resume text	Score generated	Passed
Job Fetching	Request	Jobs displayed	Passed

Table 5.4: Test Results Table

5.6 Database Representation

The Placemotor system uses **SQLite** database for storing and managing data. It is lightweight, efficient, and suitable for the current system scale. The database is designed in a structured manner with separate tables for users, resumes, analysis results, and job data.

Proper relationships are maintained using primary and foreign keys, ensuring data consistency and easy retrieval. The database supports storing user information, resume content, AI-generated results, and job listings.

Table Name	Description
User	Stores user details
Resume	Stores uploaded resumes
Analysis	Stores ATS scores and suggestions
Job	Stores job listings
Saved Jobs	Stores user saved jobs

Table 5.5: Database Tables (Overview)

Field Name	Data Type	Description
user_id	Integer (PK)	Unique user ID
email	Text	User email
password	Text	User password

Table 5.6: User Table

Field Name	Data Type	Description
resume_id	Integer (PK)	Unique resume ID
user_id	Integer (FK)	Reference to user
file	Text	Resume file path
extracted_text	Text	Extracted resume content
github_link	Text	GitHub profile link

Table 5.7: Resume Table

Field Name	Data Type	Description
analysis_id	Integer (PK)	Unique analysis ID
resume_id	Integer (FK)	Reference to resume
ats_score	Float	Resume score
suggestions	Text	Improvement suggestions

Table 5.8: Analysis Table

Field Name	Data Type	Description
job_id	Integer (PK)	Unique job ID
title	Text	Job title
company	Text	Company name
location	Text	Job location

Table 5.9: Job Table

Field Name	Data Type	Description
saved_id	Integer (PK)	Unique ID
user_id	Integer (FK)	Reference to user
job_id	Integer (FK)	Reference to job

Table 5.10: Saved Jobs Table

5.7 Discussion

The Placementor system successfully integrates AI-based resume analysis with job discovery features. It provides a user-friendly interface and delivers accurate results.

The GitHub validation feature ensures that resumes include practical work, which is essential for technical roles. The map-based job filtering enhances usability by allowing users to select preferred locations easily.

However, the system depends on external APIs and scraping sources, which may affect performance in certain situations. Despite these limitations, the system achieves its objective of assisting students in placement preparation effectively.

Feature	Benefit
AI Resume Analysis	Improves resume quality
ATS Score	Helps evaluate resumes
Job Scraper	Saves time
GitHub Validation	Encourages project work
Map Filtering	Better job matching

Table 5.11: Advantages table

Limitation	Description
AI Dependency	Output may vary
Web Scraping	Depends on external sites
Internet Requirement	Needs connectivity
Database Limitation	SQLite not scalable

Table 5.12: Limitations table

5.8 Summary

This chapter presented the results of the Placementor system, including the user interface, system modules, outputs, and database structure. All modules such as authentication, resume processing, AI analysis, job scraping, and dashboard were tested and found to be working properly.

The system successfully generates ATS scores, provides resume improvement suggestions, and displays relevant job opportunities. The database efficiently stores and manages user and system data.

Overall, the system performs effectively and meets its objective of assisting students in placement preparation.

CHAPTER 6: CONCLUSION

6.1 Introduction

This chapter summarizes the overall work carried out in the development of the Placementor system. It highlights the key achievements, evaluates the system performance, and discusses how the objectives of the project have been successfully met. The chapter also reflects on the learning outcomes and the effectiveness of the system in solving real-world problems related to placement preparation.

6.2 Summary of the Work

The Placementor system was developed as an AI-based placement assistant aimed at helping students improve their resumes and find job opportunities efficiently. The project focused on integrating modern technologies such as Artificial Intelligence, web scraping, and web-based application frameworks to build a complete solution.

The system allows users to upload resumes in PDF format, which are then processed using a PDF parser to extract textual content. This extracted content is analyzed using an AI agent built with LangChain and LLM APIs to generate ATS scores and provide suggestions for improvement.

One of the unique aspects of the system is the implementation of GitHub validation, which ensures that resumes include a GitHub profile. This encourages students to showcase practical work and technical skills, which are important in the hiring process.

The system also includes a job scraping module that collects job opportunities from platforms such as LinkedIn and other job portals. These job listings are filtered and presented to the user through a dashboard interface. The addition of a map-based location preference feature allows users to select job locations interactively, improving the relevance of job recommendations.

Other features such as Google authentication, resume templates, saved jobs, and dashboard history enhance the usability and functionality of the system.

6.3 Achievement of Objectives

The objectives defined at the beginning of the project have been successfully achieved. The system effectively performs AI-based resume analysis and generates ATS scores along with useful suggestions. It provides a platform for job discovery by fetching relevant opportunities from external sources.

The implementation of secure authentication mechanisms ensures safe access to the system. The integration of additional features such as GitHub validation, resume templates, and location-based filtering further enhances the system's capabilities.

Overall, the system meets its primary goal of assisting students in placement preparation and improving their chances of securing job opportunities.

6.4 Advantages of the System

The Placemator system offers several advantages:

- Automates resume analysis using AI
- Provides personalized suggestions for improvement
- Saves time by integrating job search and resume evaluation
- Offers real-time job opportunities from multiple platforms
- Enhances resume quality through ATS scoring
- Encourages inclusion of GitHub projects
- Provides user-friendly interface and dashboard
- Supports location-based job filtering
- Reduces manual effort in searching and applying for job opportunities
- Provides a centralized platform for resume improvement and job discovery
- Enhances user productivity by automating multiple placement-related tasks

These advantages make the system highly useful for students preparing for placements.

6.5 Limitations of the System

Despite its effectiveness, the system has certain limitations:

- The accuracy of AI-generated suggestions depends on the quality of the model
- Job scraping relies on external websites, which may change structure
- Performance may vary based on internet connectivity
- Limited dataset for training and analysis
- SQLite database may not be suitable for large-scale deployment

These limitations can be addressed in future improvements.

6.6 Learning Outcomes

The development of this project provided valuable learning experiences:

- Understanding of AI and LLM-based systems
- Experience in backend development using FastAPI
- Knowledge of frontend development using Streamlit
- Implementation of web scraping techniques
- Database design and management using SQLite
- Integration of APIs and external services
- Testing and debugging of software systems
- Understanding of API integration and external service usage
- Experience in database design and management using SQLite
- Knowledge of system architecture and modular development
- Hands-on experience in building AI-based applications
- Improved problem-solving and debugging skills

This project helped in gaining both technical and practical knowledge.

6.7 Overall Conclusion

The Placemantor system successfully demonstrates the effective use of Artificial Intelligence and modern web technologies in enhancing the placement preparation process. By integrating resume analysis, job discovery, and user-friendly features into a single platform, the system provides a comprehensive solution for students.

The application is capable of analyzing resumes, generating ATS scores, and providing meaningful suggestions to improve the quality of resumes. Additionally, the job scraping functionality enables users to access relevant job opportunities from multiple platforms, reducing the effort required in manual job searching.

The system also incorporates features such as GitHub validation, resume templates, and dashboard management, which enhance usability and ensure that users present their skills effectively. The use of a structured database ensures efficient storage and retrieval of user data and analysis results.

Overall, the project meets its intended objectives and demonstrates the potential of AI in simplifying complex processes related to recruitment and placement. With further enhancements and scalability improvements, the system can be extended into a more advanced and widely usable platform.

CHAPTER 7: FUTURE SCOPE

7.1 Introduction

This chapter discusses the possible enhancements and future improvements that can be made to the Placementor system. Although the current system successfully meets its objectives, there are several areas where the application can be extended to improve performance, scalability, and user experience.

Future developments can focus on integrating advanced technologies, expanding system capabilities, and making the platform more robust and efficient for real-world deployment.

7.2 Scope for Technical Enhancements

The current system can be improved by incorporating more advanced technical features:

- Integration of more powerful AI models to improve accuracy of resume analysis
- Implementation of deep learning techniques for better keyword matching
- Real-time processing of resumes and job data
- Improved natural language understanding for more detailed feedback

These enhancements will make the system more intelligent and reliable.

7.3 Integration with Real-Time Job APIs

Currently, the system uses web scraping to fetch job data. In the future:

- Integration with official job APIs such as LinkedIn API or other job platforms can be implemented
- Real-time job updates can be provided
- More accurate and structured job data can be retrieved

This will improve reliability and reduce dependency on scraping.

7.4 Advanced Personalization

The system can be enhanced to provide more personalized user experiences:

- Personalized job recommendations based on user profile and skills
- AI-based career guidance and suggestions
- Skill gap analysis and learning recommendations
- Custom resume suggestions for different job roles

This will make the system more user-centric and effective.

7.5 Mobile Application Development

The current system is web-based. In the future:

- A mobile application can be developed for Android and iOS
- Users can access the system anytime and anywhere
- Push notifications for job alerts can be added

This will increase accessibility and user engagement.

7.6 Enhanced Resume Builder

The resume template feature can be expanded further:

- More professional and customizable templates
- AI-assisted resume generation
- Real-time editing and preview
- Export options in multiple formats

This will improve usability and flexibility.

7.7 Improved Security Features

Security can be enhanced by:

- Implementing advanced authentication methods
- Encrypting user data
- Secure storage of sensitive information
- Multi-factor authentication This will ensure better protection of user data.

7.8 Scalability Improvements

To support a larger number of users:

- Migration from SQLite to scalable databases such as PostgreSQL
- Deployment on cloud platforms
- Load balancing and performance optimization

This will make the system suitable for large-scale use.

7.9 Addition of Admin Module

An admin panel can be introduced to:

- Monitor system performance
- Manage users and data
- Control job listings and updates
- Analyze system usage

This will improve system management and control.

7.10 Integration with Learning Platforms

The system can be extended to include:

- Integration with online learning platforms
- Course recommendations based on skill gaps
- Certification tracking

This will help users improve their skills along with job preparation.

7.11 Overall Future Vision

In the future, Placemotor can evolve into a complete career assistance platform that not only helps in resume building and job search but also supports skill development, career planning, and professional growth. The system can also be extended using advanced AI models and cloud-based deployment to support large-scale real-world applications.

APPENDIX I: ADDITIONAL DETAILS AND SYSTEM OUTPUTS

A.1 Introduction

This appendix contains additional information related to the Placemotor system, including sample outputs, API details, and supporting data used during the development and testing of the system. These details provide further insight into the working of the application.

A.2 Sample Resume Input

The system accepts resumes in PDF format. A sample resume used for testing includes the following information:

- Name and contact details
- Educational qualifications
- Technical skills
- Project experience
- Certifications

This data is extracted and used as input for AI-based analysis.

A.3 Sample ATS Analysis Output

The system generates an ATS score along with detailed feedback. A typical output includes:

- ATS Score (percentage)
- Strengths of the resume
- Weaknesses and missing keywords
- Suggestions for improvement

This helps users improve their resumes for better job opportunities.

A.4 Sample Job Data

The system fetches job listings from external platforms such as LinkedIn and other sources. The job data includes:

- Job title
- Company name

- Location
- Job description

These results are displayed based on user preferences and resume content.

A.5 API Endpoints Used

The backend system uses FastAPI to handle requests. The main API endpoints include:

- /login – Handles user authentication
- /upload-resume – Uploads resume files
- /analyze-resume – Performs AI-based resume analysis
- /fetch-jobs – Retrieves job listings

These APIs enable communication between frontend and backend.

A.6 Tools and Technologies

The system is developed using the following technologies:

- Streamlit – Frontend interface
- FastAPI – Backend development
- SQLite – Database management
- LangChain and LLM API – AI-based analysis
- BeautifulSoup – Job scraping

A.7 Additional Notes

The system is designed to be scalable and can be further enhanced by integrating advanced AI models, improving job filtering, and expanding database capabilities.

APPENDIX II: SYSTEM IMPLEMENTATION DETAILS

B.1 Introduction

This appendix provides additional technical details related to the implementation of the Placementor system. It includes information about system workflow, database structure, and integration of various components used in the application.

B.2 System Workflow Description

The Placementor system follows a structured workflow for processing user inputs and generating outputs:

1. User logs into the system
2. Resume is uploaded in PDF format
3. Text is extracted from the resume
4. AI model analyzes the content
5. ATS score and suggestions are generated
6. Job listings are fetched from external sources
7. Results are displayed to the user

This workflow ensures smooth interaction between different system modules.

B.3 Database Schema Overview

The system uses SQLite database to store user and application data. The main tables include:

- User Table – Stores user credentials
- Resume Table – Stores uploaded resumes
- Analysis Table – Stores ATS scores and feedback
- Job Table – Stores job listings
- Saved Jobs Table – Stores user-selected jobs

The database ensures efficient storage and retrieval of data.

B.4 Integration of AI Module

The system integrates an AI-based analysis module using LangChain and LLM APIs. This module performs:

- Resume parsing
- Keyword extraction
- ATS score calculation
- Suggestion generation

The AI module enhances the system's capability to provide intelligent feedback.

B.5 Job Scraping Mechanism

The job scraper module collects job data from external platforms such as LinkedIn and other sources. It processes job listings and filters them based on user preferences and resume content.

B.6 Limitations of Implementation

- The system depends on external APIs and internet connectivity
- AI-generated outputs may vary in accuracy
- Web scraping results may change based on source availability

B.7 Future Improvements

The system can be enhanced by:

- Adding real-time job updates
- Improving AI accuracy
- Integrating cloud-based databases
- Developing a mobile application

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